

Final Exam of Algorithms and Data Structures 2023-2024

An industrial bakery wants to improve its order management system, and tasks you with developing a software simulation of the bakery's operations. The entire simulation runs in discrete time. Assume that one time instant elapses after each input command is processed. The simulation starts at time 0. The following elements must be considered in the simulation:

- The **ingredients** of the sweets, each identified by its *name*, a sequence of characters.
- The set of **recipes** offered by the bakery, also identified by a *name*. Each recipe uses different *quantities* of each required ingredient (indicated by an integer, in grams).
- The **warehouse** of bakery ingredients, which stocks every ingredient in use. The warehouse is **restocked** by new **batches** of ingredients, according to a schedule set by the supplier. Each **batch** is characterised by a *quantity* (always in grams) and an *expiry date*, indicated by the number of the time instant from which the batch is expired.
- Customers place **orders** for one or more sweets via an online platform or by phone. In either case, the bakery immediately proceeds to prepare the ordered sweets. The advanced (and expensive) machinery that prepares sweets is so fast that preparation of an arbitrary number of sweets can be assumed to occur in a single time instant of the simulation. The ingredients needed for each preparation are taken from the warehouse by always preferring batches with the nearest expiry date. If insufficient ingredients are available to allow the preparation *in full* of an order, it is put *on hold*. Any number of orders may be on hold simultaneously. The bakery proceeds to prepare any subsequent orders. Upon each restocking, the bakery evaluates whether, with the newly received ingredients, it can prepare orders currently on hold. If so, it prepares them within the same time instant. Pending orders are fulfilled in chronological order of *arrival* of the order.
- Periodically, the **courier** visits the bakery to collect completed orders. Upon arrival, orders to be loaded are selected in chronological order of arrival. The process stops as soon as an order is encountered that would exceed the remaining capacity (in grams) on the truck. Assume that the weight of each prepared sweet equals the sum of the gram quantities of each ingredient. Each order is always loaded as a whole. Once the orders are selected, the bakery loads them in decreasing order of weight. Ties in weight are broken by chronological order of arrival.

Assume that all quantities are integers greater than zero, regardless of unit. Names, whether of ingredients or recipes, are defined over the alphabet $\{a, \dots, z, A, \dots, Z, 0, \dots, 9, _\}$ and are at most 255 characters long. The simulation ends after the last command is read.

The input file begins with a line containing two integers: the courier period and its capacity. It is followed by a sequence of commands, one per line, in the following format. All positive or null integer values fit in 32 bits.

- `aggiungi_ricetta` $\langle recipe_name \rangle$ $\langle ingredient_name \rangle$ $\langle quantity \rangle$...
Example: `aggiungi_ricetta meringhe_della_prozia zucchero 100 albumi 100`
Adds a recipe to the catalogue. The number of $(\langle ingredient_name \rangle \langle quantity \rangle)$ pairs is arbitrary. If a recipe with the same name is already present, it is ignored.
Expected output: `aggiunta` or `ignorato`.
- `rimuovi_ricetta` $\langle recipe_name \rangle$
Example: `rimuovi_ricetta cannoncini`
Removes a recipe from the catalogue. No effect if the recipe is not present, or if there are orders related to it that have not yet been shipped.
Expected output: `rimossa`, `ordini in sospeso` or `non presente`.
- `rifornimento` $\langle ingredient_name \rangle$ $\langle quantity \rangle$ $\langle expiry \rangle$...
Example: `rifornimento zucchero 200 150 farina 1000 220`
The bakery is restocked with a set of batches, one per ingredient. The number of batches is arbitrary.
Expected output: `rifornito`.
- `ordine` $\langle recipe_name \rangle$ $\langle item_count \rangle$
Example: `ordine torta_paradiso 36`
Places an order for $\langle item_count \rangle$ sweets with recipe $\langle recipe_name \rangle$.
Expected output: `accettato` or `rifiutato` if no recipe with the specified name exists.

In addition to the outputs listed above, the program prints the orders loaded onto the courier's truck as a sequence of triples $\langle order_arrival_time \rangle \langle recipe_name \rangle \langle item_count \rangle$, one per line, in loading order. For a courier of period n , the output is produced before processing the commands at time kn with $k \in 1, 2, \dots$. If the truck is empty, the message `camioncino vuoto` is printed.

Example

The following example illustrates the expected behaviour of the program. The column *time* indicates the current time instant when each command is executed.

<i>t</i>	Input	Output	Comment
–	5 325		Courier configuration
0	aggiungi_ricetta torta farina 50 uova 10 zucchero 20	aggiunta	Recipe added
1	aggiungi_ricetta ciambella farina 20 uova 5 burro 2	aggiunta	
2	aggiungi_ricetta profiterole farina 10 uova 2 latte 3 zucchero 3 cioccolato 4	aggiunta	
3	rimuovi_ricetta sfogliatella	non presente	Recipe does not exist
4	rifornimento farina 100 10 uova 100 10 zucchero 100 10 burro 100 10 latte 100 10 cioccolato 100 10	rifornito	
5	ordine ciambella 6	camioncino vuoto accettato	First shipment, at time 5, before reading the next command Not enough flour; order put on hold
6	ordine profiterole 3	accettato	Order prepared and immediately queued for shipment
7	rimuovi_ricetta profiterole	ordini in sospeso	The <i>profiterole</i> orders are queued for shipment
8	aggiungi_ricetta pane_dolce farina 1 zucchero 1 uova 1	aggiunta	
9	ordine ciambella 3	accettato	Order prepared and immediately queued for shipment
10		9 ciambella 3 6 profiterole 3	The ciambella order is heavier (81g vs. 66g) and is loaded first
	ordine torta 1	accettato	The pantry is empty because the batches from the last restock are expired: the order goes on hold
11	rifornimento farina 100 15 farina 50 13 uova 45 20 zucchero 20 20 burro 15 20	rifornito	The ciambella order from time 5 is prepared because it has been waiting longest; the torta ordered at time 10 remains on hold
12	rifornimento farina 100 15 uova 7 15 zucchero 25 15 latte 5 15 cioccolato 5 15	rifornito	This third restock unblocks the torta order
13	ordine torta 1	accettato	
14	ordine profiterole 1	accettato	Torta and <i>profiterole</i> are prepared immediately
15		5 ciambella 6 10 torta 1 13 torta 1	The <i>profiterole</i> orders remain queued because they do not fit on the truck and are the most recent order. The two torte have the same weight, so they are loaded in chronological order. What remains of the batches expiring at time 15 is removed from the pantry.

At the end of execution, 3 g of butter and 2 g of sugar remain unused. The *profiterole* order at time 14 is never shipped.